A Few Notes on Vector Calculus

David Meyer

dmm@1-4-5.net

Last update: March 16, 2022

1 Introduction

TBD

2 Notation

Some of the notation we'll use is described in this section.

2.1 Vectors

Some authors use bold font to denote a vector, e.g. $\mathbf{u}$. Adding a "hat" denotes a unit vector. That is, for a vector $\mathbf{u}$, $\hat{\mathbf{u}}$ is defined to be

$$\hat{\mathbf{u}} := \frac{\mathbf{u}}{\|\mathbf{u}\|}$$

In words: $\hat{\mathbf{u}}$ is a vector of unit length in the $\mathbf{u}$ direction.

Similarly, $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{i}}$ are unit vectors in $\mathbb{R}^3$ in the x, y, and z directions respectively. Note that $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ are the canonical basis vectors for $\mathbb{R}^3$ [2] and have column vector format

$$\hat{\mathbf{i}} = \begin{bmatrix}1\\0\\0\end{bmatrix},\ \hat{\mathbf{j}} = \begin{bmatrix}0\\1\\0\end{bmatrix},\ \hat{\mathbf{k}} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

I have also seen $\mathbf{e}_i$ used to represent the ith basis vector in $\mathbb{R}^n$. So $\mathbf{e}_i$ is a vector with a one in the ith position and a zero in each of the other $n-1$ positions. In $\mathbb{R}^3$ this means that $\mathbf{e}_1 = \hat{\mathbf{i}}$, $\mathbf{e}_2 = \hat{\mathbf{j}}$, and $\mathbf{e}_3 = \hat{\mathbf{k}}$.

In general, the standard basis (which is sometimes called the computational basis) for the n-dimensional Euclidean space consists of the ordered set of n distinct vectors

$$\{\mathbf{e}_i : 1 \leq i \leq n\}$$

where $\mathbf{e}_i$ is the ith basis vector, that is, it has a one in the ith coordinate (position) and zeros everywhere else.[1] The $\mathbf{e}_i$ have column vector format

[1] This is also called a "one-hot" encoding in machine learning parlance, where the i's might be the classes in a classification problem.

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \ldots, \mathbf{e}_{n-1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

Using these definitions we can define the parametric form of some curve C in $\mathbb{R}^3$, $\mathbf{r}(t)$, as follows:

$$\mathbf{r}(t) = g(t)\hat{\mathbf{i}} + h(t)\hat{\mathbf{j}} + k(t)\hat{\mathbf{k}}$$

where $t \in [a, b]$ and $g(t)$, $h(t)$, and $k(t)$ are scalar functions[2].

Another common notation for vectors is $\vec{r}(t)$, where $\vec{r}(t)$ might be defined as follows:

$$\vec{r}(t) = g(t)\hat{i} + h(t)\hat{j} + k(t)\hat{k}$$

In this document I'll use these notations interchangeably.

3 Definitions and a Few Theorems

Definition 3.1. Vector Notation:

Here we will use $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ to represent a row or column vector in some n dimensional space (usually $\mathbb{R}$ or $\mathbb{C}$). If $\mathbf{a}$ is a column vector then in matrix format

$$\mathbf{a} = \underbrace{\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}}_{n \times 1}$$

Alternatively, if $\mathbf{a}$ is a row vector then in matrix format

$$\mathbf{a} = \underbrace{\begin{bmatrix} a_1, a_2, \ldots, a_n \end{bmatrix}}_{1 \times n}$$

The *transpose* of a vector $\mathbf{a}$, $\mathbf{a}^\intercal$, is defined as follows. If $\mathbf{a}$ is a column vector then

$$\mathbf{a}^\intercal = \begin{bmatrix} a_1, a_2, \ldots, a_n \end{bmatrix}$$

Alternatively, if $\mathbf{a}$ is a row vector then

$$\mathbf{a}^\intercal = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

[2] A scalar function is a function f such that $f : \mathbb{R}^n \to \mathbb{R}$.

Definition 3.2. Dot Product: The dot product (aka scalar product or inner product) of two n-dimensional vectors $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$ is defined to be the scalar value

$$\mathbf{a} \cdot \mathbf{b} := \langle \mathbf{a}, \mathbf{b} \rangle := \sum_{j=1}^{n} a_j b_j$$

If $\mathbf{a}$ and $\mathbf{b}$ are column vectors then the dot product can also be written as a matrix product: $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^\intercal \mathbf{b}$ where $\mathbf{a}^\intercal$ is the transpose of the vector $\mathbf{a}$. Similarly, if $\mathbf{a}$ and $\mathbf{b}$ are row vectors then the dot product can be written as $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}\mathbf{b}^\intercal$.

The (matrix) shape of the dot product of two $n \times 1$ column vectors $\mathbf{a}$ and $\mathbf{b}$ is

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^\intercal \mathbf{b} \to \underbrace{\begin{bmatrix} \cdots\cdots\cdots \end{bmatrix}}_{1 \times n} \underbrace{\begin{bmatrix} \cdots \\ \vdots \\ \cdots \end{bmatrix}}_{n \times 1} \to c$$

where $c \in \mathbb{R}$.

According to the Pythagorean theorem, the length of a vector $\mathbf{a} = (a_1, a_2, a_3) \in \mathbb{R}^3$, denoted $\|\mathbf{a}\|$, equals $\sqrt{a_1^2 + a_2^2 + a_3^2}$. The next definition is a generalization of the notion of length to vectors in $\mathbb{R}^n$.

Definition 3.3. Euclidean Norm: The Euclidean norm of a vector

$$\mathbf{x} = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$$

is defined to be

$$\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{\sum_{j=1}^{n} x_j^2} \tag{1}$$

Definition 3.4. p-Norm: The p-norm of a vector

$$\mathbf{x} = (x_1, x_2 \ldots, x_n)$$

for $p \geq 1, p \in \mathbb{R}$. The p-norm (also called ℓ_p-norm) $\|\mathbf{x}\|_p$ is defined to be

$$\|\mathbf{x}\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

For $p = 1$, we get the taxicab norm, for $p = 2$ we get the Euclidean norm, and as p approaches ∞ the p-norm approaches the infinity norm or maximum norm: $\|\mathbf{x}\|_\infty := \max_i |x_i|$.

Next, assume that $\mathbf{x}$ and $\mathbf{y}$ are two linearly independent vectors in $\mathbb{R}^3$ and that M is the plane spanned by them. Then these two vectors generate a triangle in M with sides of length $\|\mathbf{x}\|$, $\|\mathbf{y}\|$, and $\|\mathbf{x}-\mathbf{y}\|$. If $\theta \in \{0, \pi\}$ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$ in M, then the Law of Cosines [6] gives us

$$\|\mathbf{x}-\mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\cos(\theta)\|\mathbf{x}\|\|\mathbf{y}\|$$

We can also see that

$$\begin{aligned}
\|\mathbf{x}-\mathbf{y}\|^2 &= \langle \mathbf{x}-\mathbf{y}, \mathbf{x}-\mathbf{y}\rangle && \text{\# by Equation (1)} \\
&= \sum_{j=1}^{n} (x_j - y_j)^2 && \text{\# by Equation (1)} \\
&= \sum_{j=1}^{n} (x_i^2 - 2x_j y_j + y_j^2) && \text{\# expand previous line} \\
&= \sum_{j=1}^{n} x_j^2 - 2\sum_{j=1}^{n} x_j y_j + \sum_{j=1}^{n} y_j && \text{\# sum is a linear operator} \\
&= \|\mathbf{x}\|^2 + -2\langle \mathbf{x}, \mathbf{y}\rangle + \|\mathbf{y}\|^2 && \text{\# by Equation (1)} \\
&= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\langle \mathbf{x}, \mathbf{y}\rangle && \text{\# rearrange}
\end{aligned}$$

Combining these expressions for $\|\mathbf{x}-\mathbf{y}\|^2$ gives us

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\cos(\theta)\|\mathbf{x}\|\|\mathbf{y}\| = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\langle \mathbf{x}, \mathbf{y}\rangle$$

Subtracting $\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$ from both sides, dividing by -2, and rearranging gives us

$$\langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x}\|\|\mathbf{y}\|\cos\theta \tag{2}$$

Definition 3.5. Orthogonal Vectors: $\langle \mathbf{x}, \mathbf{y}\rangle = 0$

We say that $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are orthogonal if $\langle \mathbf{x}, \mathbf{y}\rangle = 0$, since

$$\begin{aligned}
\langle \mathbf{x}, \mathbf{y}\rangle &= \|\mathbf{x}\|\|\mathbf{y}\|\cos\theta && \text{\# by Equation (2)} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x}\|\|\mathbf{y}\|\cos\tfrac{\pi}{2} && \text{\# } \mathbf{x} \text{ and } \mathbf{y} \text{ orthogonal} \Rightarrow \theta = \tfrac{\pi}{2} \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x}\|\|\mathbf{y}\|\,0 && \text{\# } \cos\tfrac{\pi}{2} = 0 \\
&\Rightarrow \langle \mathbf{x}, \mathbf{y}\rangle = 0 && \text{\# } \|\mathbf{x}\|\|\mathbf{y}\|\,0 = 0
\end{aligned}$$

Theorem 3.1. Cauchy–Schwarz Inequality: $\left|\langle \mathbf{a}, \mathbf{b}\rangle\right| \leq \|\mathbf{a}\|\|\mathbf{b}\|$

Proof: The inequality is trivial if either $\mathbf{a}$ or $\mathbf{b}$ is zero, so assume that neither is. So without loss of generality let $\mathbf{x} = \dfrac{\mathbf{a}}{\|\mathbf{a}\|}$ and $\mathbf{y} = \dfrac{\mathbf{b}}{\|\mathbf{b}\|}$ and so $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ ($\mathbf{x} = \hat{\mathbf{a}}$ and $\mathbf{y} = \hat{\mathbf{b}}$). Then

$$
\begin{aligned}
0 &\leq \|\mathbf{x}-\mathbf{y}\| && \text{\# } \|\mathbf{x}-\mathbf{y}\| \geq 0 \text{ for all } \mathbf{x},\mathbf{y} \in \mathbb{R}^n \text{ by Equation (1)} \\
&\leq \|\mathbf{x}-\mathbf{y}\|^2 && \text{\# } \|\mathbf{x}-\mathbf{y}\| \geq 0 \Rightarrow \|\mathbf{x}-\mathbf{y}\|^2 \geq 0 \\
&\leq \sum_{j=1}^{n}(x_j-y_j)^2 && \text{\# } \|\mathbf{x}-\mathbf{y}\|^2 = \sum_{j=1}^{n}(x_j-y_j)^2 \text{ by Equation (1)} \\
&\leq \sum_{j=1}^{n}\left(x_j^2 - 2x_jy_j + y_j^2\right) && \text{\# expand previous line} \\
&\leq \sum_{j=1}^{n}x_j^2 - 2\sum_{j=1}^{n}x_jy_j + \sum_{j=1}^{n}y_j^2 && \text{\# sum is a linear operator} \\
&\leq \|\mathbf{x}\|^2 - 2\langle\mathbf{x},\mathbf{y}\rangle + \|\mathbf{y}\|^2 && \text{\# } \|\mathbf{x}\|^2 = \sum_{j=1}^{n}x_j^2 \text{ and } \langle\mathbf{x},\mathbf{y}\rangle = \sum_{j=1}^{n}x_jy_j \text{ by Equation (1)} \\
&\leq 1 - 2\langle\mathbf{x},\mathbf{y}\rangle + 1 && \text{\# } \|\mathbf{x}\| = \|\mathbf{y}\| = 1 \Rightarrow \|\mathbf{x}\|^2 = \|\mathbf{y}\|^2 = 1 \\
&\leq 2 - 2\langle\mathbf{x},\mathbf{y}\rangle && \text{\# } 1+1=2 \\
&\Rightarrow -2 \leq -2\langle\mathbf{x},\mathbf{y}\rangle && \text{\# subtract 2 from both sides} \\
&\Rightarrow 1 \geq \langle\mathbf{x},\mathbf{y}\rangle && \text{\# divide by -2, reversing inequality} \\
&\Rightarrow \langle\mathbf{x},\mathbf{y}\rangle \leq 1 && \text{\# rearrange} \\
&\Rightarrow \left\langle \frac{\mathbf{a}}{\|\mathbf{a}\|}, \frac{\mathbf{b}}{\|\mathbf{b}\|} \right\rangle \leq 1 && \text{\# } \mathbf{x} = \frac{\mathbf{a}}{\|\mathbf{a}\|},\ \mathbf{y} = \frac{\mathbf{b}}{\|\mathbf{b}\|} \text{ by definition above} \\
&\Rightarrow \frac{\langle\mathbf{a},\mathbf{b}\rangle}{\|\mathbf{a}\|\|\mathbf{b}\|} \leq 1 && \text{\# scalar multiplication rule: } \langle c_1\mathbf{v}_1, c_2\mathbf{v}_2\rangle = c_1c_2\langle\mathbf{v}_1,\mathbf{v}_2\rangle \\
&\Rightarrow \langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\| && \text{\# multiply both sides by } \|\mathbf{a}\|\|\mathbf{b}\|
\end{aligned}
$$

If we replace $\mathbf{a}$ by $-\mathbf{a}$ we see by Equation (1) that $\|-\mathbf{a}\| = \|\mathbf{a}\|$. Further, by the scalar multiplication rule for dot products with $c_1 = -1$ and $c_2 = 1$ we have $\langle -\mathbf{a},\mathbf{b}\rangle = -\langle\mathbf{a},\mathbf{b}\rangle$. Taken together these results imply that $-\langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$.

Next note that the absolute value of x, $|x|$, is defined to be

$$
|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases} \tag{3}
$$

So one way to see the Cauchy–Schwarz inequality is by setting x equal to $\langle\mathbf{a},\mathbf{b}\rangle$ in Equation (3). Then when $\langle\mathbf{a},\mathbf{b}\rangle \geq 0$ we have $\left|\langle\mathbf{a},\mathbf{b}\rangle\right| = \langle\mathbf{a},\mathbf{b}\rangle$ and we showed above that $\langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Similarily, when $\langle\mathbf{a},\mathbf{b}\rangle < 0$ we have $\left|\langle\mathbf{a},\mathbf{b}\rangle\right| = -\langle\mathbf{a},\mathbf{b}\rangle$ and we showed that $-\langle\mathbf{a},\mathbf{b}\rangle \leq \|\mathbf{a}\|\|\mathbf{b}\|$. Together these give us $\left|\langle\mathbf{a},\mathbf{b}\rangle\right| \leq \|\mathbf{a}\|\|\mathbf{b}\|$, namely, the Cauchy–Schwarz inequality. $\square$

3.1 Derivatives

We will also denote the derivative of a vector $\mathbf{r}$ with respect to t by $\dot{\mathbf{r}}$, $\dfrac{d\mathbf{r}}{dt}$ or $\mathbf{r}'(t)$. Note also that by definition

$$
\frac{d\mathbf{r}}{dt} := \lim_{\Delta t \to 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t}
$$

3.2 Vector Fields

In vector calculus and physics, a vector field is an assignment of a vector to each point in a subset of space [3]. For example, a vector field in the plane can be visualized as a collection of arrows with a given magnitude and direction, each attached to a point in the plane. Vector fields are often used to model the speed and direction of a moving fluid throughout space, or the strength and direction of some force, such as the magnetic or gravitational force, as it changes from one point to another point.

For example, the wind velocity vector field for the 2011 Joplin, MO tornado [7] is shown in Figure 1. Here the color represents the wind speed $\|\mathbf{v}\|$, where $\mathbf{v}$ is the wind velocity vector.

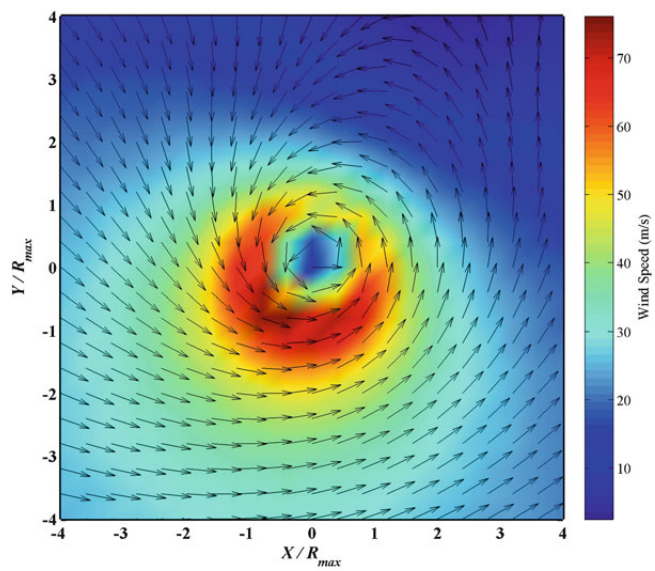

Figure 1: 2011 Joplin, MO Tornado Wind Velocity Vector Field

The general form of a vector field (here in three dimensions) is

$$\vec{F}(x, y, z) = P(x, y, z)\,\hat{i} + Q(x, y, z)\,\hat{j} + R(x, y, z)\,\hat{k}$$

where P, Q, and R are scalar functions.

4 Arc Length

Consider a segment of a parametric curve $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}} + k(t)\,\hat{\mathbf{k}}$ between two points $P = \mathbf{r}(t)$ and $Q = \mathbf{r}(t + \Delta t)$, as shown in Figure 2.

The next question we might ask is what is the length of the segment of the curve between P and Q? This length is called the arc length and is denoted by Δs and can be approximated by the chord length $\|\Delta\mathbf{r}\|$.

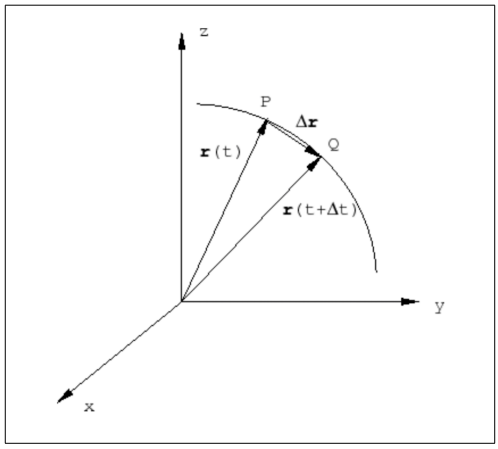

Figure 2: $\Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$

Specifically we can see from Figure 2 that

$$\|\Delta r\| = \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\|$$

and so

$$\Delta s \approx \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\|$$

Now we can compute the infinitesimal arc length ds:

$$\begin{aligned}
\Delta s &\approx \|\Delta\mathbf{r}\| && \text{\# approximate } \Delta s \text{ by } \|\Delta\mathbf{r}\| \\
&= \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\| && \text{\# } \Delta\mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t) \\
&= \|\mathbf{r}(t + \Delta t) - \mathbf{r}(t)\| \cdot \frac{\Delta t}{\Delta t} && \text{\# multiply by } 1 = \frac{\Delta t}{\Delta t} \\
&= \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# assume } \Delta t > 0 \\
\Rightarrow\ & \Delta s \approx \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t && \text{\# combine expressions for } \Delta s \\
\Rightarrow\ & \lim_{\Delta t \to 0} \Delta s = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# take the limit of both sides} \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left[\left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \Delta t \right] && \text{\# } \Delta t \to 0 \Rightarrow \Delta s \to 0 \Rightarrow \lim_{\Delta t \to 0} \Delta s = ds \\
\Rightarrow\ & ds = \lim_{\Delta t \to 0} \left\| \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# product rule for limits [1]} \\
\Rightarrow\ & ds = \left\| \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \right\| \cdot \lim_{\Delta t \to 0} \Delta t && \text{\# for a normed vector space } X \ \lim \|x_n\| = \|\lim x_n\| \\
\Rightarrow\ & ds = \left\| \frac{d\mathbf{r}}{dt} \right\| \cdot dt && \text{\# } \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{d\mathbf{r}}{dt},\ \lim_{\Delta t \to 0} \Delta t = dt
\end{aligned}$$

Interestingly we can also see this using the Pythagorean theorem. Specifically, if we move an infinitesimal distance in the x and y directions we get a triangle in the x-y plane who's sides have length dx and dy and who's hypotenuse is ds. This scenario is shown for some curve $f(x)$ in Figure 3.

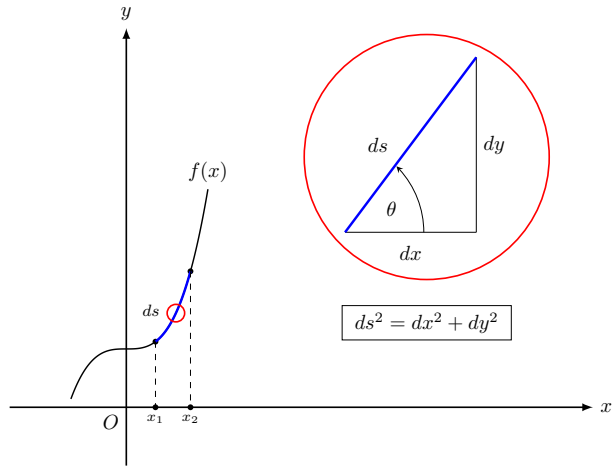

Figure 3: $f(x)$, ds and the Pythagorean Theorem

In the two dimensional case where the parameterization our $x = g(t)$ and $y = h(t)$ we have $\mathbf{r}(t) = g(t)\,\hat{\mathbf{i}} + h(t)\,\hat{\mathbf{j}}$. Further, since derivative is a linear operator [8] we know that

$$\frac{d\mathbf{r}}{dt} = \frac{dg}{dt}\,\hat{\mathbf{i}} + \frac{dh}{dt}\,\hat{\mathbf{j}} \tag{4}$$

The Pythagorean theorem tells us that $ds = \sqrt{dx^2 + dy^2}$. Using this expression for ds we can see that

$$\begin{aligned}
ds &= \sqrt{dx^2 + dy^2} && \text{\# Figure 3} \\
&= \sqrt{dg^2 + dh^2} && \text{\# switch to parametric form: } x = g(t) \text{ and } y = h(t) \\
&= \sqrt{dg^2 + dh^2} \cdot \frac{dt}{dt} && \text{\# multiply by } 1 = \frac{dt}{dt},\ dt > 0 \\
&= \sqrt{dg^2 + dh^2} \cdot \left[\sqrt{\frac{1}{dt^2}} \cdot dt\right] && \text{\# } \frac{x}{x} = \sqrt{\left(\frac{1}{x}\right)^2} \cdot x = \sqrt{\frac{1}{x^2}} \cdot x,\ x > 0 \\
&= \sqrt{\left(dg^2 + dh^2\right) \cdot \left(\frac{1}{dt^2}\right)} \cdot dt && \text{\# simplify} \\
&= \sqrt{\left(\frac{dg}{dt}\right)^2 + \left(\frac{dh}{dt}\right)^2} \cdot dt && \text{\# simplify} \\
&= \sqrt{g'(t)^2 + h'(t)^2} \cdot dt && \text{\# switch notation from Leibniz} \rightarrow \text{Lagrange: } \frac{df}{dt} = f'(t) \\
&= \left\|\frac{d\mathbf{r}}{dt}\right\| \cdot dt && \text{\# Equation (4) and the definition of } \|\cdot\|
\end{aligned}$$

Now we can see that

$$\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\|$$

and so

$$\frac{ds}{dt} = \|\dot{\mathbf{r}}\| \tag{5}$$

4.1 The Unit Tangent Vector T

Since $\frac{ds}{dt} = \left\| \frac{d\mathbf{r}}{dt} \right\| = \|\dot{\mathbf{r}}\|$ we also know that the unit tangent vector $\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} = \frac{d\mathbf{r}}{ds}$ [4]. One way to see this is to notice that

$$\begin{aligned}
\mathbf{T} &= \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} && \text{\# definition of the unit tangent vector } \mathbf{T} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\|\dot{\mathbf{r}}\|} && \text{\# } \dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} \text{ (definition of } \dot{\mathbf{r}}\text{)} \\
&= \frac{\left(\frac{d\mathbf{r}}{dt}\right)}{\left(\frac{ds}{dt}\right)} && \text{\# } \|\dot{\mathbf{r}}\| = \frac{ds}{dt} \text{ (Equation (5))} \\
&= \frac{d\mathbf{r}}{ds} && \text{\# simplify}
\end{aligned}$$

5 Line Intergals

6 Surface Integrals

7 Conclusions

8 Acknowledgements

Thanks to Dave Neary who pointed out that using the reverse triangle inequality in my proof of the Cauchy–Schwarz inequality was overkill. In particular, $\|\mathbf{x} - \mathbf{y}\| \geq 0$ for all $\mathbf{x}$ and $\mathbf{y}$ by Equation (1), so the triangle inequality (or reverse triangle inequality) was not needed.

9 Appendix A

I had thought that one way to prove the Cauchy–Schwarz inequality is to use the reverse triangle inequality, but Dave Neary pointed out that I didn't need it. So I'm leaving the derivation of the reverse triangle inequality in this appendix.

Theorem 9.1. Reverse Triangle Inequality: $\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x} - \mathbf{y}\|$

We can derive the reverse triangle inequality from the triangle inequality [5, 9] by observing that

$$\begin{aligned}
\|\mathbf{x}\| &= \|\mathbf{x} + (-\mathbf{y} + \mathbf{y})\| && \text{\# add 0 to } \mathbf{x}\text{: } \mathbf{x} = \mathbf{x} + 0 = \mathbf{x} + (-\mathbf{y} + \mathbf{y}) \\
&= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| && \text{\# addition is associative: } \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3) = (\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 \\
&\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| && \text{\# by the triangle inequality: } \|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\| \\
&\Rightarrow \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| && \text{\# combine expressions for } \|\mathbf{x}\| \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| && \text{\# subtract } \|\mathbf{y}\| \text{ from both sides}
\end{aligned}$$

and

$$\begin{aligned}
\|\mathbf{y}\| &= \|\mathbf{y} + (-\mathbf{x} + \mathbf{x})\| && \text{\# add 0 to } \mathbf{y}\text{: } \mathbf{y} = \mathbf{y} + 0 = \mathbf{y} + (-\mathbf{x} + \mathbf{x}) \\
&= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| && \text{\# addition is associative} \\
&\leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| && \text{\# by the triangle inequality} \\
&\Rightarrow \|\mathbf{y}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| && \text{\# combine expressions for } \|\mathbf{y}\| \\
&\Rightarrow \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| && \text{\# subtract } \|\mathbf{x}\| \text{ from both sides} \\
&\Rightarrow -\Big[\|\mathbf{y}\| - \|\mathbf{x}\|\Big] \geq -\|\mathbf{y} - \mathbf{x}\| && \text{\# multiply both sides by -1: } -1 * (\mathbf{v}_1 \leq \mathbf{v}_2) \Rightarrow -\mathbf{v}_1 \geq -\mathbf{v}_2 \\
&\Rightarrow \|\mathbf{x}\| - \|\mathbf{y}\| \geq -\|\mathbf{x} - \mathbf{y}\| && \text{\# } \|\mathbf{y} - \mathbf{x}\| = \|-(\mathbf{y} - \mathbf{x})\| = \|\mathbf{x} - \mathbf{y}\| \text{ by Equation (1)} \\
&\Rightarrow -\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| && \text{\# rearrange}
\end{aligned}$$

Combining the expressions for $\|\mathbf{x}\| - \|\mathbf{y}\|$ we get $-\|\mathbf{x} - \mathbf{y}\| \leq \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|$, that is, the reverse triangle inequality $\big|\|\mathbf{x}\| - \|\mathbf{y}\|\big| \leq \|\mathbf{x} - \mathbf{y}\|$. $\square$

LaTeX Source

References

[1] Alex Svirin. Properties of Limits. `https://math24.net/properties-limits.html`. [Online; accessed 14-November-2021].

[2] David Meyer. Notes on Change of Basis. `https://davidmeyer.github.io/qc/change_of_basis.pdf`, January 2019. [Online; accessed 14-March-2022].

[3] Galbis, A. and Maestre, M. *Vector Analysis Versus Vector Calculus*. Universitext. Springer New York, 2012.

[4] Gregory Hartman. Unit Tangent and Normal Vectors. `https://sites.und.edu/timothy.prescott/apex/web/apex.Ch12.S4.html`, Nov 2021. [Online; accessed 08-March-2022].

[5] Jean H. Gallier. Vector Norms and Matrix Norms. `https://www.cis.upenn.edu/~cis515/cis515-11-sl4.pdf`, 2020. [Online; accessed 10-March-2022].

[6] Math Open Reference. Proof of the Law of Cosines. `https://www.mathopenref.com/lawofcosinesproof.html`, 2011. [Online; accessed 16-March-2022].

[7] NIST. Joplin Missouri Tornado 2011. https://www.nist.gov/disaster-failure-studies/joplin-tornado-ncst-investigation, Aug 2011. [Online; accessed 10-March-2022].

[8] Wikipedia contributors. Linearity of Differentiation — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Linearity_of_differentiation&oldid=1026177205, 2021. [Online; accessed 1-September-2021].

[9] Wikipedia Contributors. Triangle Inequality — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Triangle_inequality&oldid=1074965160, 2022. [Online; accessed 10-March-2022].